

Document 3: Information Architecture & Wireframe Spec

System architecture, endpoint routing, and component-level interface designs for the AI-Powered Governance Intelligence Platform.

1. Sitemap & URL Routing Architecture

Route	Access Scope	Interface Description & Core Capabilities
/	Public ▾	System landing and authentication entry redirect point.
/auth/login	Public ▾	Multi-factor Government SSO, ID.gov.in integration, and Supabase Auth.
/auth/unauthorized	Public ▾	Access Denied view triggered by insufficient LGD or role claims.
/app	Authenticated ▾	Layout wrapper enforcing RBAC and active session token validation.
/app/overview	All Roles ▾	Executive Command Center featuring national choropleth and top-level KPI metrics.
/app/convergence	Central Exec, Dept Nodal, ... ▾	Cross-scheme intelligence hub.
/app/convergence/matrix	Central Exec, Dept Nodal, ... ▾	Cross-scheme overlap and gap analysis matrix.
/app/convergence/compare	Central Exec, Dept Nodal, ... ▾	Side-by-side multi-scheme metric comparison tool.
/app/geo	All Roles ▾	Geographic intelligence spatial views.

Route	Access Scope	Interface Description & Core Capabilities
/app/geo/national	All Roles ▾	National macro choropleth rendering cross-state metrics.
/app/geo/state/[stateLgd]	All Roles ▾	State-level district drilldown and spatial distribution.
/app/geo/district/[distLgd]	All Roles ▾	District block-level convergence canvas and localized map layers.
/app/schemes	All Roles ▾	Ingested scheme administrative views.
/app/schemes/directory	All Roles ▾	Comprehensive catalog of all ingested national and state schemes.
/app/schemes/[schemeId]	All Roles ▾	Granular scheme drilldown (MGNREGA, PM-KISAN, PMAY-G, etc.).
/app/intelligence	All Roles ▾	AI reasoning and anomaly feeds.
/app/intelligence/query	All Roles ▾	Natural language query and reasoning workbench.
/app/intelligence/alerts	All Roles ▾	Anomaly detection engine and stalled project alert feed.
/app/admin	Super Admin ▾	System administration and data pipeline controls.
/app/admin/pipelines	Super Admin ▾	Ingestion scheduler, ETL error logs, and LGD mapping status.
/app/admin/users	Super Admin ▾	User provisioning and LGD scope assignment management.

2. Screen-by-Screen Component Inventory

Screen 1: Executive Command Center (/app/overview)

Header Navigation

- **Left Cluster:** Government Emblem, Platform Brand Identifier ("Samanvay Intelligence"), Active Environment Status Badge.
- **Center Cluster:** Global Natural Language Search Prompt Bar ("Ask a cross-scheme question...").
- **Right Cluster:** Fiscal Year Selector (FY 2025-26), LGD Scope Chip Selector (All India), User Profile & Role Indicator.

Section 1: Executive KPI Metric Strip

- **Card 1: Total Outlay vs. Expenditure**
 - Primary metric display: ₹ Cr Utilized vs. ₹ Cr Allocated.
 - Visual indicator: Circular progress ring showing overall percentage utilization.
- **Card 2: Active Beneficiary Footprint**
 - Primary metric display: Unique De-duplicated Individual Beneficiaries.
 - Demographic breakdown: Sub-text distribution for SC, ST, and Women categories.
- **Card 3: Cross-Scheme Interventions**
 - Primary metric display: Total Active Works Across Rural Development + Housing.
 - Contextual change indicator relative to the previous fiscal quarter.
- **Card 4: System Critical Anomalies**
 - Primary metric display: Count of Stalled or Divergent Districts.
 - Visual indicator: High-priority critical alert pill with direct filter trigger.

Section 2: Interactive Geographic Intelligence

- **Left Column (8-Span): Map Canvas**
 - Toolbar: View controls including Scheme Convergence, Fund Utilization, and Gap Index.
 - Interactive Vector Map: High-resolution India vector canvas with GeoJSON hover tooltips for key metrics.
- **Right Column (4-Span): Divergence Panel**
 - Panel Header: "Top Divergent Districts" (Fund Flow High / Milestone Lagging).
 - District List Component: Dynamic repeating list showing District Name, State, Deviation Score, and "Drill Down" action link.

Section 3: Recent Cross-Ministry Intelligence Insights

- **Section Header:** "AI-Detected Implementation Bottlenecks" (Auto-generated from nightly pipelines).

- **3-Column Insight Grid:**
 - Insight Card A (Overlap): PMAY-G construction delayed in 14 blocks due to timber transport embargo.
 - Insight Card B (Gap): 32,000 PM-KISAN landholders missing MGNREGA job card linkages in Bundelkhand.
 - Insight Card C (Velocity): Jal Jeevan Mission fund utilization surging 4x faster than asset validation.

Screen 2: Natural Language Query & Reasoning Workbench (/app/intelligence/query)

Left Rail: Query Composer & History (5-Span)

- **Preset Prompt Chips:** One-click sample queries (e.g., "Show fund stagnation > 60 days", "Compare PMAY-G vs MGNREGA in tribal districts").
- **NLP Query Input Textarea:** Auto-expanding text input featuring token count display and active schema hint pills.
- **Execution Controls:** "Generate Insight" action button alongside a Confidence Strictness slider.
- **Recent Audit Queries List:** Interactive list of past user sessions displaying query text, timestamp, and query latency metrics.

Right Workspace: Dynamic Intelligence Response Canvas (7-Span)

- **State Management Container:** Visual rendering states for Empty, Loading Skeleton, and Active Content.
- **Response Header:** Echoed Formulated Question with "Verified Against Schema" execution status badge.
- **AI Synthesized Answer Card:** Structured natural-language narrative summary with key data points highlighted in bold anchors.
- **Dynamic Data Visualization:** Context-aware chart generation (Grouped Bar, Multi-line Trend, or District Heatmap based on SQL response structure).
- **Evidence & Provenance Accordion (Traceability Layer):**
 - Tab 1 (Generated SQL Query): Read-only syntax-highlighted SQL with execution time in milliseconds.
 - Tab 2 (Source Tables & Joins): Visual schema breadcrumb chain (e.g., `mord_mgnrega_perf` ⋈ `moa_pmkisan`).
 - Tab 3 (Raw Tabular Data Output): Paginated, sortable interactive data table with CSV and XLSX export triggers.